Akoizo Original Practice - Form 1

Division B | Solar System | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which three broad needs define habitability for life as we know it?

A. Liquid water, energy, and key chemical elements

B. Oxygen, sunlight, and soil

C. A magnetic field, rings, and moons

D. Ice, methane, and nitrogen only

Answer: _____

2. (2 points) Why is Venus's surface hotter than Mercury's?

Unit: Venus and Mars: rocky neighbors.
Worked-case data: A Mars orbiter detects clay minerals in layered rocks at the end of a channel that opens into a crater, forming a fan shape. What can you conclude?
Separate investigation observations:
Orbital image: A fan of layered sediment where a valley enters a crater.
Spectra: Clay minerals and carbonates detected in the fan.
Rover data: Fine-grained mudstone layers on the crater floor.

A. A thick CO_2 atmosphere creates a runaway greenhouse

B. Venus is closer to the Sun

C. Venus has a strong magnetic field

D. Venus has active tidal heating

Answer: _____

3. (2 points) Which method directly gives a planet's radius?

Unit: Exoplanets: detection and the named systems.
Worked-case data: An Earth-sized planet transits two stars: a Sun-like star and a red dwarf with one-tenth the Sun's radius. Compare the transit depths.
Separate investigation observations:
Light curve: Seven different periodic dips in brightness.
Star: An ultracool red dwarf, about 40 light-years away.
Sizes: All planets are roughly Earth-sized.
JWST: The two innermost planets show no thick atmosphere.

A. Transit

B. Radial velocity

C. Microlensing alone

D. Astrometry

Answer: _____

4. (2 points) If the distance between two masses triples, the gravitational force becomes:

A. 3 times as large

B. 1/27 as large

C. 1/9 as large

D. 1/3 as large

Answer: _____

5. (2 points) Which class of molecules forms cell membranes?

Unit: Small bodies and the building blocks of life.
Worked-case data: Earth's oceans have a D/H ratio of about 1.6 x 10□^4. Comet 67P's water has about 5.3 x 10□^4, and carbon-rich asteroids have values close to Earth's. Which source better matches Earth's water?
Separate investigation observations:
Sample minerals: Clay minerals and sodium carbonate salts.
Organics: Amino acids and nucleobases detected in the lab.
Isotopes: D/H ratio close to Earth's oceans.

A. Nucleic acids

B. Carbohydrates

C. Lipids

D. Proteins

Answer: _____

6. (2 points) A cool object near 300 K emits most strongly in:

A. Infrared

B. Visible

C. Ultraviolet

D. X-rays

Answer: _____

7. (2 points) What liquids fill Titan’s lakes and seas?

Unit: Ocean worlds: Europa, Ganymede, Enceladus, and Titan.
Worked-case data: A spacecraft flying by a moon detects a magnetic field that changes direction as the moon moves through its planet’s field. What does this suggest about the moon’s interior?
Separate investigation observations:
Atmosphere: Thick nitrogen, 1.5 times Earth’s surface pressure.
Surface temperature: About -180 °C.
Radar: Smooth, dark lakes near the poles.
Probe images: Rounded pebbles on the landing site.

A. Methane and ethane

B. Liquid water

C. Sulfuric acid

D. Liquid nitrogen only

Answer: _____

8. (4 points) Explain why being in the habitable zone does not guarantee liquid water.

Answer: _____
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_____

9. (4 points) Explain why liquid water is not stable on Mars’s surface today.

Unit: Venus and Mars: rocky neighbors.
Worked-case data: A Mars orbiter detects clay minerals in layered rocks at the end of a channel that opens into a crater, forming a fan shape. What can you conclude?
Separate investigation observations:
Orbital image: A fan of layered sediment where a valley enters a crater.
Spectra: Clay minerals and carbonates detected in the fan.
Rover data: Fine-grained mudstone layers on the crater floor.

Answer: _____
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10. (4 points) Explain how combining transit and radial velocity data reveals a planet's composition.

Unit: Exoplanets: detection and the named systems.
Worked-case data: An Earth-sized planet transits two stars: a Sun-like star and a red dwarf with one-tenth the Sun's radius. Compare the transit depths.
Separate investigation observations:
Light curve: Seven different periodic dips in brightness.
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Sizes: All planets are roughly Earth-sized.
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Answer: _____
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11. (4 points) Why does the Moon cause larger tides on Earth than the Sun?

Unit: Gravity, orbits, and tides.
Worked-case data: Planet A has 4 times Earth's mass and 2 times Earth's radius. Compare its surface gravity and escape velocity with Earth's.
Separate investigation observations:
Orbit: Moon B orbits its planet in 3.6 days on a slightly elliptical path.
Rotation: It always shows the same face to the planet.
Neighbors: Its period is exactly twice that of an inner moon.
Surface: Young, crack-covered ice with few craters.

Answer: _____
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12. (4 points) What is a clathrate, and where might it matter?

Unit: Small bodies and the building blocks of life.
Worked-case data: Earth's oceans have a D/H ratio of about 1.6 x 10□^4. Comet 67P's water has about 5.3 x 10□^4, and carbon-rich asteroids have values close to Earth's. Which source better matches Earth's water?
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Organics: Amino acids and nucleobases detected in the lab.
Isotopes: D/H ratio close to Earth's oceans.

Answer: _____
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13. (4 points) Star A is 6,000 K and star B is 3,000 K. Compare their peak wavelengths.

Answer: _____
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14. (4 points) Compare Titan’s methane cycle with Earth’s water cycle.

Unit: Ocean worlds: Europa, Ganymede, Enceladus, and Titan.
Worked-case data: A spacecraft flying by a moon detects a magnetic field that changes direction as the moon moves through its planet’s field. What does this suggest about the moon’s interior?
Separate investigation observations:
Atmosphere: Thick nitrogen, 1.5 times Earth’s surface pressure.
Surface temperature: About -180 °C.
Radar: Smooth, dark lakes near the poles.
Probe images: Rounded pebbles on the landing site.

Answer: _____
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15. (4 points) Why would oxygen and methane together in an exoplanet atmosphere be interesting?

Answer: _____
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16. (4 points) Why did Perseverance land in Jezero crater?

Unit: Venus and Mars: rocky neighbors.
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Separate investigation observations:
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Spectra: Clay minerals and carbonates detected in the fan.
Rover data: Fine-grained mudstone layers on the crater floor.

Answer: _____
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17. (4 points) Why is direct imaging easiest for young, large planets far from their stars?

Answer: _____
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18. (6 points) Solve this case and explain your reasoning: Planet A has 4 times Earth’s mass and 2 times Earth’s radius. Compare its surface gravity and escape velocity with Earth’s.

Answer: _____
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